

VANESA AGUAY GUERRA

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EDUCATION

Yale University, *BS in Electrical Engineering & Computer Science*, New Haven, CT May, 2028
Relevant Coursework: Data Structures & Programming; Data Analysis; Linear Algebra; Intermediate Microeconomics; Electronics.

AT&T, *Technology Academy*, (Azure cloud, data analytics, REST APIs, AI/ML, cybersecurity) May 2025-Aug. 2025

EXPERIENCE

Yale Information Technology Services, *Student Technician (STC)*, New Haven, CT Oct. 2025-Current
• Deliver campus-wide tech support, troubleshooting hardware, software, and network issues through collaborative problem-solving.

Yale Engineering, *Quantum Networking Researcher*, New Haven, CT Sept. 2025-Current
• Integrated real-time measurement systems with sub-microsecond synchronization, improving quantum-laser experiment timing accuracy by > 95% for the Quantum Laser Across the Sound team.

OpenAI, *ChatGPT Lab Member* Oct. 2025-Current
• Collaborating with ~50 students nationwide synthesizing AI learning use cases shared with 1M+ students globally to improve educational engagement.

Tsai Center for Innovative Thinking at Yale (Tsai CITY), *Fellow*, New Haven, CT Sept. 2025-Current
• Collaborated with 6 cross-disciplinary teams on venture development, aligning project resources and stakeholder goals.
• Conducted data-driven initiatives as 1 of 10 selected fellows across Yale College and the graduate schools.

NASA L'SPACE, *Proposal Writing & Evaluation Program and Mission Concepts Academy* May 2025-Current
• Developed hardware architecture development for a \$450M Mars rover mission, working with 17 engineers and improving subsystem power-budget efficiency by ~15% through load-balancing and array-controller optimization.
• Conducted parametric cost and risk analysis using NASA CEH 4.0, reducing projected mission risk by ~8% through balanced thermal and power subsystem integration.

Yale La Casa Latino Cultural Center, *Peer Liaison*, New Haven, CT Mar. 2024-Current
• Conducted data analysis on engagement across 21 mentees increasing first-year participation retention by 90%.
• Led digital and in-person initiatives reaching 190+ students nationwide, including La Casa's largest recruitment events series.

Yale School of the Environment, *Website & Communications Manager*, New Haven, CT Aug. 2024-May 2025
• Managed site with 400+ posts and 50+ pages, implementing editorial workflows that cut update time by ~50% for student op-eds and sustainability research content.

Jane Street, *Focus and Unboxed Student*, New York, NY May 2024 & May 2025
• Applied quantitative analysis using SQL and probabilistic modeling to collaboratively solve trading problems achieving 1st place.

Teaching Assistant, *Code Haven (Yale-New Haven Public Schools)* Dec. 2024- May 2025
• Taught weekly Python classes to 30+ students and helped run an 80-student hackathon, increasing class project completion by 60%.

PROJECTS

Algorithmic Systems Engineering | C, Probability Modeling, Data Structures, Performance Profiling
• Built hash tables and KD-trees in C with dynamic resizing and amortized O(1) lookup, passing 100+ unit and Valgrind tests.
• Developed a probabilistic Blotto battle simulator, optimizing traversal algorithms for a 40% runtime speedup.

xDrip+ ML Calibration | Python, TensorFlow
• Initiating an open-source collaboration applying machine learning to improve glucose sensor calibration and real-time prediction accuracy for diabetes management.

NOTABLE HONORS

- Hispanic Scholarship Fund Scholar, HSF (<5% from 69k nationally), 2025
- National NCWIT AiC Winner, *Bank of America* and *National Center for Women & Info Technology* (1/39 nationally), 2024

PUBLICATIONS

"Governing Space Security: Analysis of the Potential for a Binding Multilateral Treaty" [American Institute of Aeronautics & Astronautics](#)
Hola Mundo: Inteligencia Artificial para Ninos (bilingual AI literacy book) [Amazon](#)

SKILLS & INTERESTS

Programming: Python, C, Java, SQL, Racket, C++

Tools: Siemens NX, MATLAB, Arduino, Altium, TensorFlow, Jupyter, Azure

Interests: Drone videography (DJI Spark) and post-production in Final Cut Pro, AI ethics, and half-marathon running.